Amazon ML Challenge 2026

Complete Video Content — Business Entity Resolution

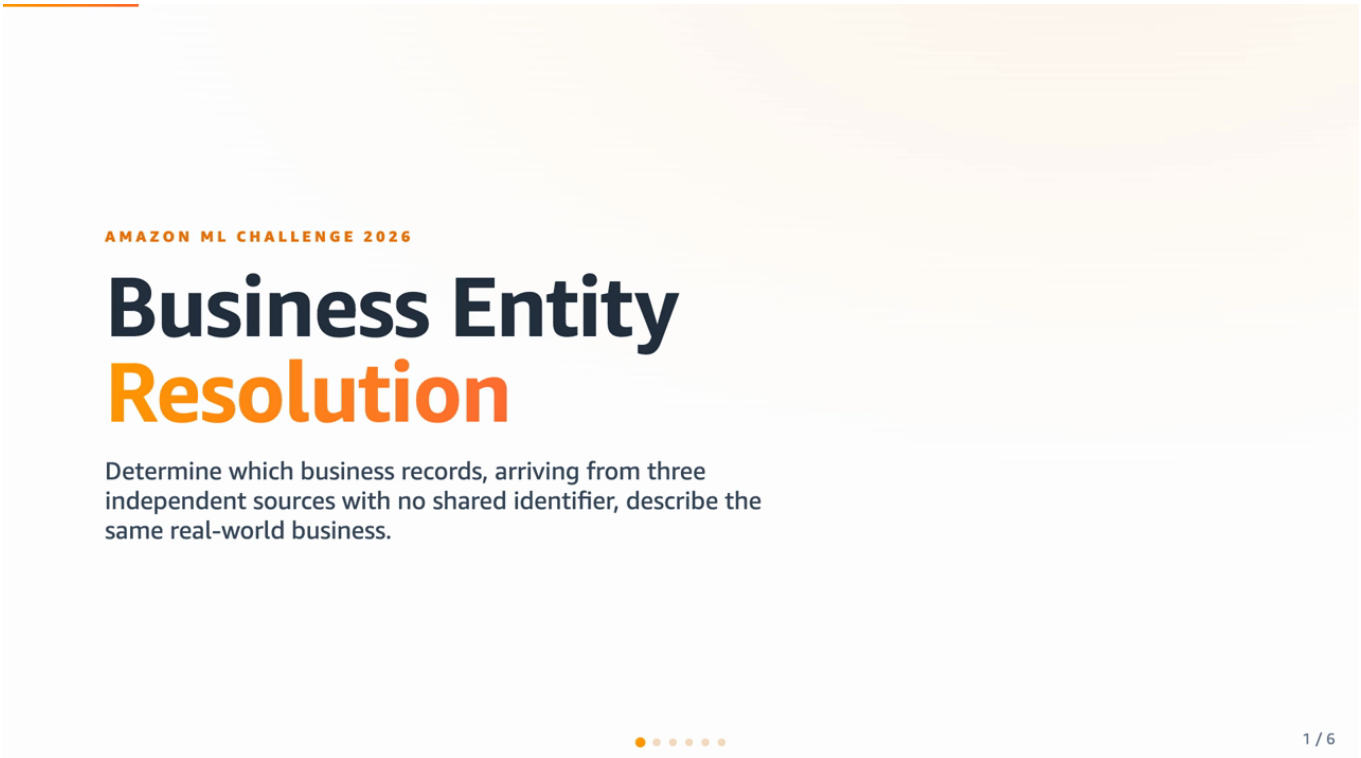
Full written content with representative screenshots from the supplied video. The text is preserved rather than reduced to a short summary.

COMPLETE VIDEO CONTENT — BUSINESS ENTITY RESOLUTION

NOTE

This document is a faithful, expanded transcription of the information presented on the six slides/sections of the supplied video. It preserves the examples, identifiers, file names, relationships, submission structure, scoring formula, and challenge tips rather than reducing them to a short summary.

SLIDE 1 OF 6 — BUSINESS ENTITY RESOLUTION

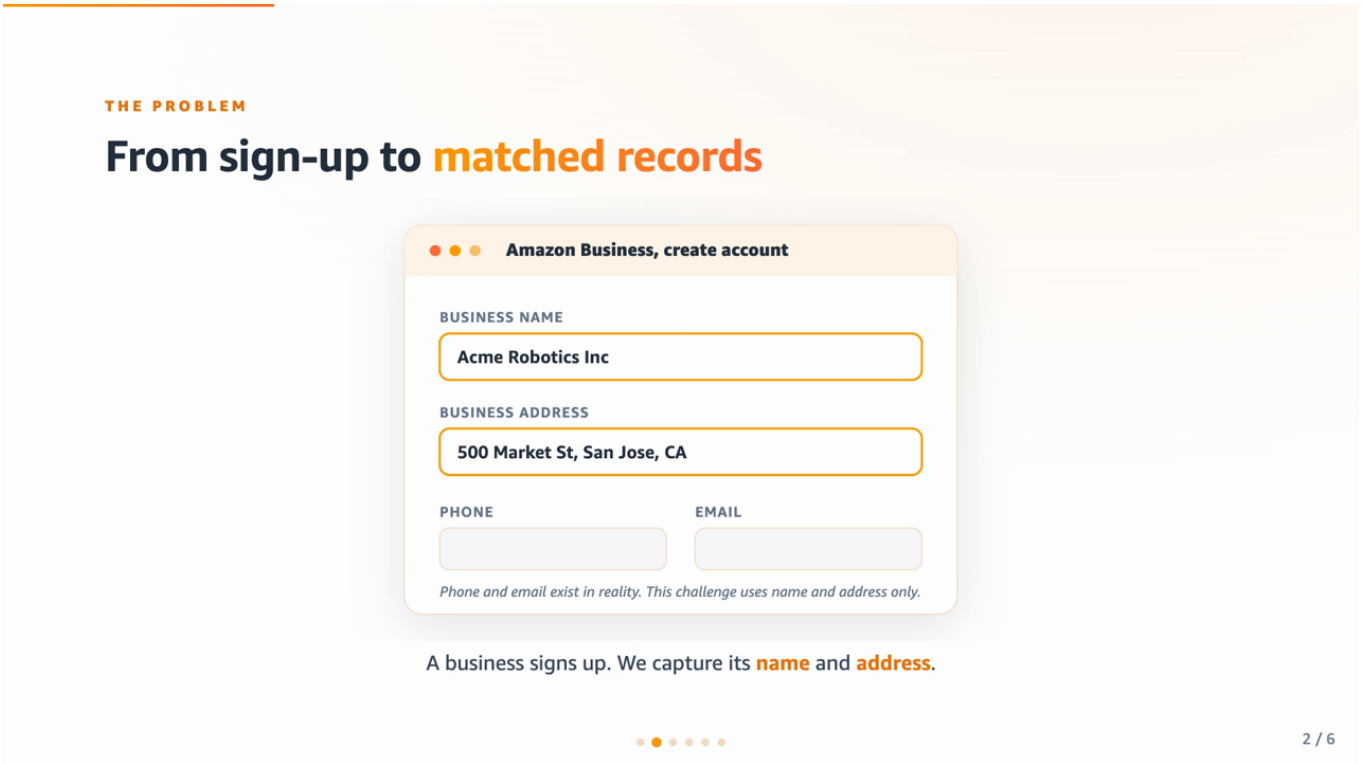


AMAZON ML CHALLENGE 2026

Business Entity Resolution

Determine which business records, arriving from three independent sources with no shared identifier, describe the same real-world business.

SLIDE 2 OF 6 — THE PROBLEM



THE PROBLEM

From sign-up to matched records

A business signs up. We capture its name and address.

The sign-up screen/record contains:

BUSINESS NAME

BUSINESS ADDRESS

PHONE

EMAIL

Phone and email exist in reality. This challenge uses name and address only.

ONE BUSINESS, THREE REPRESENTATIONS

The same real-world business can appear in three independent data sources.

SOURCE 1

Name: Acme Robotics Inc.

Address: 500 Market St, San Jose

SOURCE 2

Name: Acme Robotics Incorporated (or, in the later tabular example, Acme Robotics Inc)

Address: 500 Market Street, San Jose

SOURCE 3

Name: Acme Robotics

Address: Nr. City Hall, San Jose

The slide explains:

One business, three representations, each from a different data vendor, with no shared ID.

ENTITY RESOLUTION

The objective is to connect the Source 1 entity to the records from Source 2 and Source 3 that refer to the same real-world business.

For the Acme Robotics example:

• Source 2 contains a true match.

• Source 3 contains a true match.

• There can also be records that look similar but are not the same business.

The example explicitly shows:

SOURCE 2, match SOURCE 3, match SOURCE 2, look-alike Acme Bakery LLC (look-alike example)

Entity resolution links every Source 1 entity to its matches: zero, one, or many.

BLOCKING / CANDIDATE GENERATION

The video then shows a larger example involving three sources.

SOURCE 1

`S1-732914 — Acme Robotics Inc`

500 Market St, San Jose

`S1-889301 — Delta Foods`

8 Oak Ave, Austin

`S1-410562 — Bright Cafe LLC`

22 Pine St, Reno

`S1-205774 — Zen Traders`

4 Hill Rd, Boise

SOURCE 2

`S2-118820 — Acme Robotics Inc`

500 Market Street, San Jose

`S2-540221 — Acme Robotix`

12 Elm Rd, San Jose

`S2-397155 — Delta Foods Co`

8 Oak Avenue, Austin

`S2-663049 — Bright Cafe`

22 Pine Street, Reno

SOURCE 3

`S3-905477 — Acme Robotics`

Nr. City Hall, San Jose

`S3-063118 — Acme Bakery`

500 Market St, San Jose

`S3-651230 — Delta Foods Ltd`

Oak Ave, Austin

`S3-472088 — Kappa Motors`

90 Lake Dr, Fargo

BLOCKS SHOWN IN THE EXAMPLE

Each block is keyed on name + address.

The slide visually groups the records into:

Acme Delta Bright Cafe

The Acme block is shown as an example.

`candidate_pairs.tsv`

For S1-732914, the candidate list is:

```
S1-732914 maps to:
S2-118820
S2-540221
S3-905477
S3-063118
```

This is described as:

one block, one shortlist

The purpose of blocking is stated explicitly:

Blocking buckets records by a cheap key built from name and address, so similar names or shared addresses group together, matches and look-alikes alike.

Important distinction: Blocking does NOT decide which candidates are true matches. It creates a shortlist of records that should be considered by the matching model.

MATCHING AFTER BLOCKING

The video next shows the candidate list being filtered.

For S1-732914:

Candidate:

`S2-118820 — match`

Candidate:

`S3-905477 — match`

Candidate:

`S2-540221 — look-alike`

Candidate:

S3-063118 — look-alike

The model filters candidates and keeps only true matches.

The resulting file is:

matching_results.tsv

Example final result:

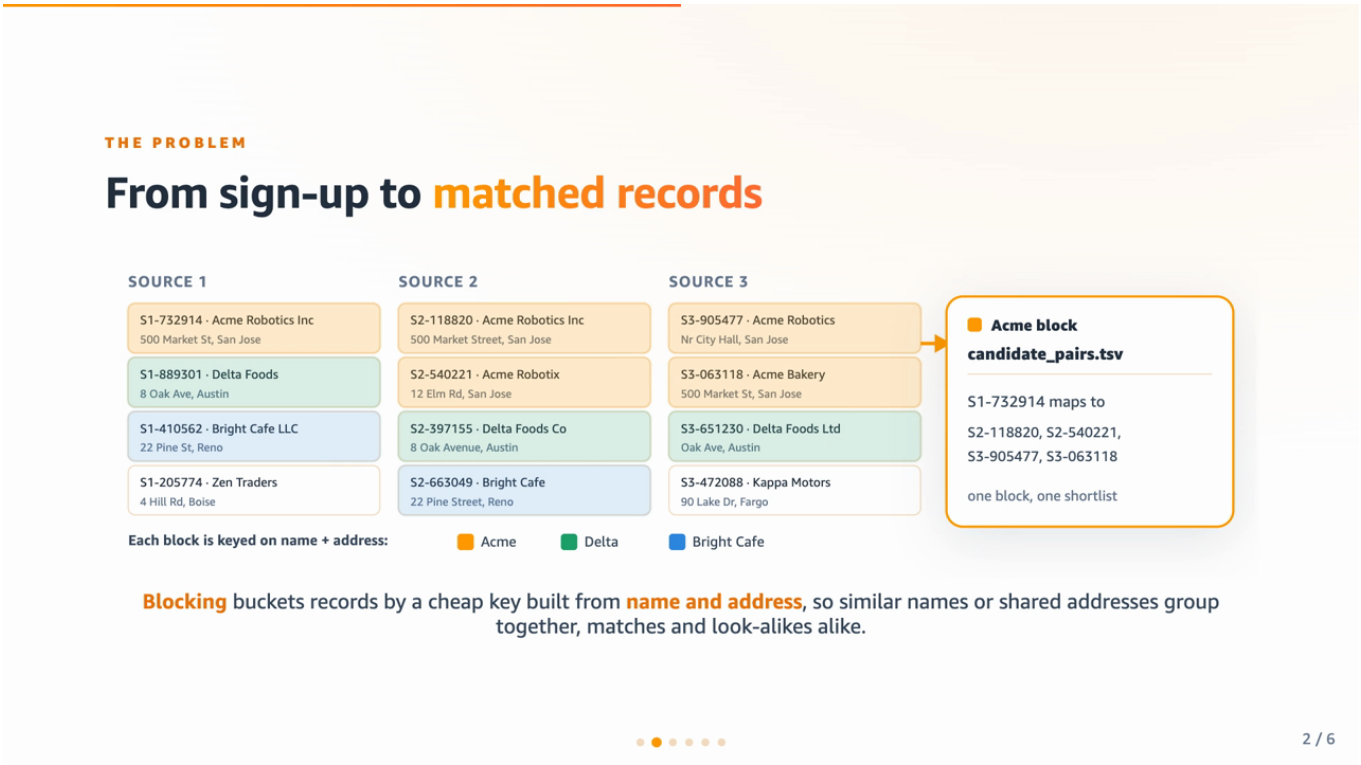
S1-732914 -> S2-118820, S3-905477

The slide explicitly says:

one row per Source 1 entity

Thus, candidate_pairs.tsv represents the blocking/candidate stage, while matching_results.tsv represents the final entity-resolution matches.

SLIDE 3 OF 6 — THE DATASET



THE DATASET

Train to learn, test to prove it

TRAINING SET

The training set contains:

• Records across all 3 sources.

• Ground-truth labels: train_ground_truth.tsv

• Use it to build and evaluate your model.

TEST SET

The test set contains:

• The same 3 sources, with no labels.

• Produce a prediction for every Source 1 entity.

• This is what the leaderboard scores.

GROUND-TRUTH FORMAT

Ground truth is one row per Source 1 entity.

Its Source 1 ID maps to a comma-separated list of all matching Source 2 and Source 3 IDs.

Therefore, a single row holds the entity's full match set.

If a Source 1 entity matches nothing, the list is empty.

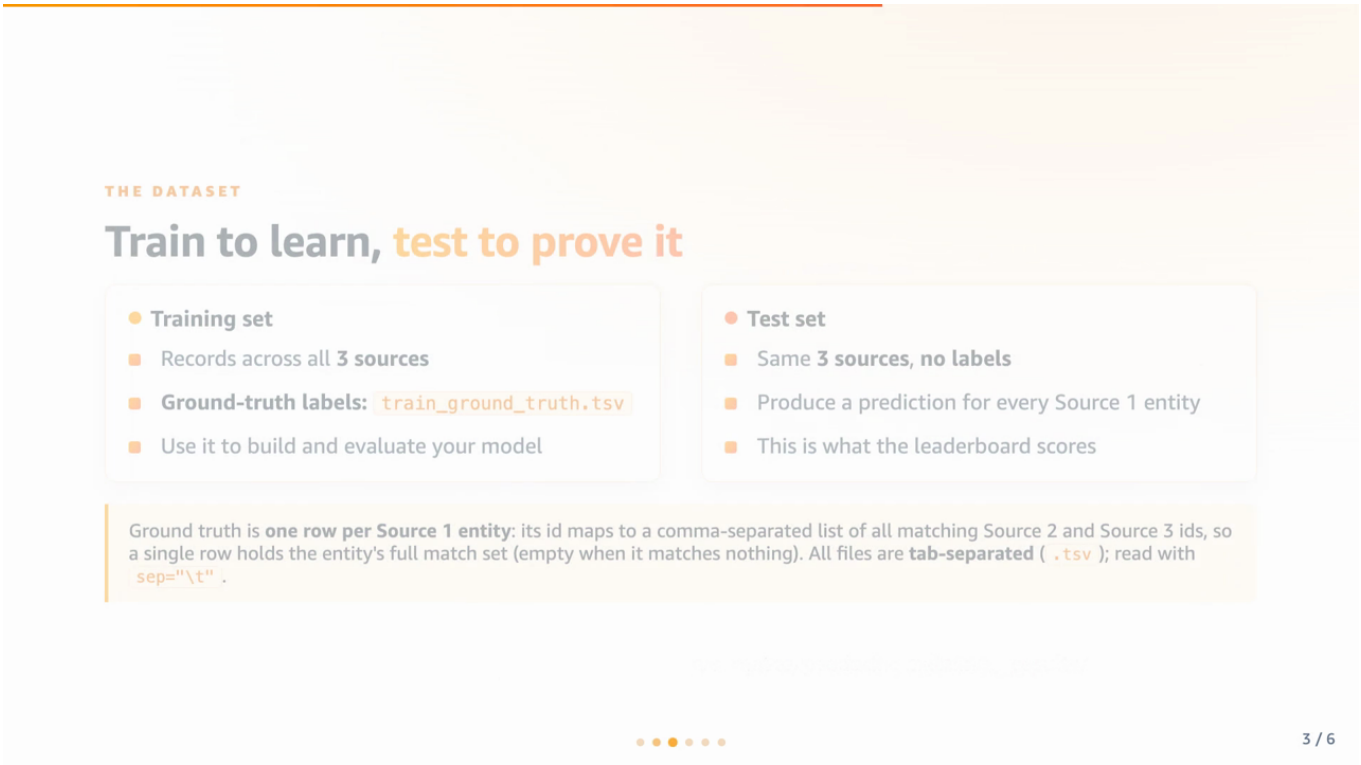
FILE FORMAT

All files are tab-separated (.tsv).

They should be read using:

```
sep="\\t"
```

SLIDE 4 OF 6 — WHAT YOU SUBMIT



WHAT YOU SUBMIT

A file, then a package

DURING THE CHALLENGE

Upload:

`matching_results.tsv`

This is the only file scored on the leaderboard.

There is one row per Source 1 entity.

An empty list means no match.

FINAL PACKAGE (.zip)

The final ZIP package contains:

output/matching_results.tsv final matches

output/candidate_pairs.tsv blocking set (audited)

code/... runnable pipeline

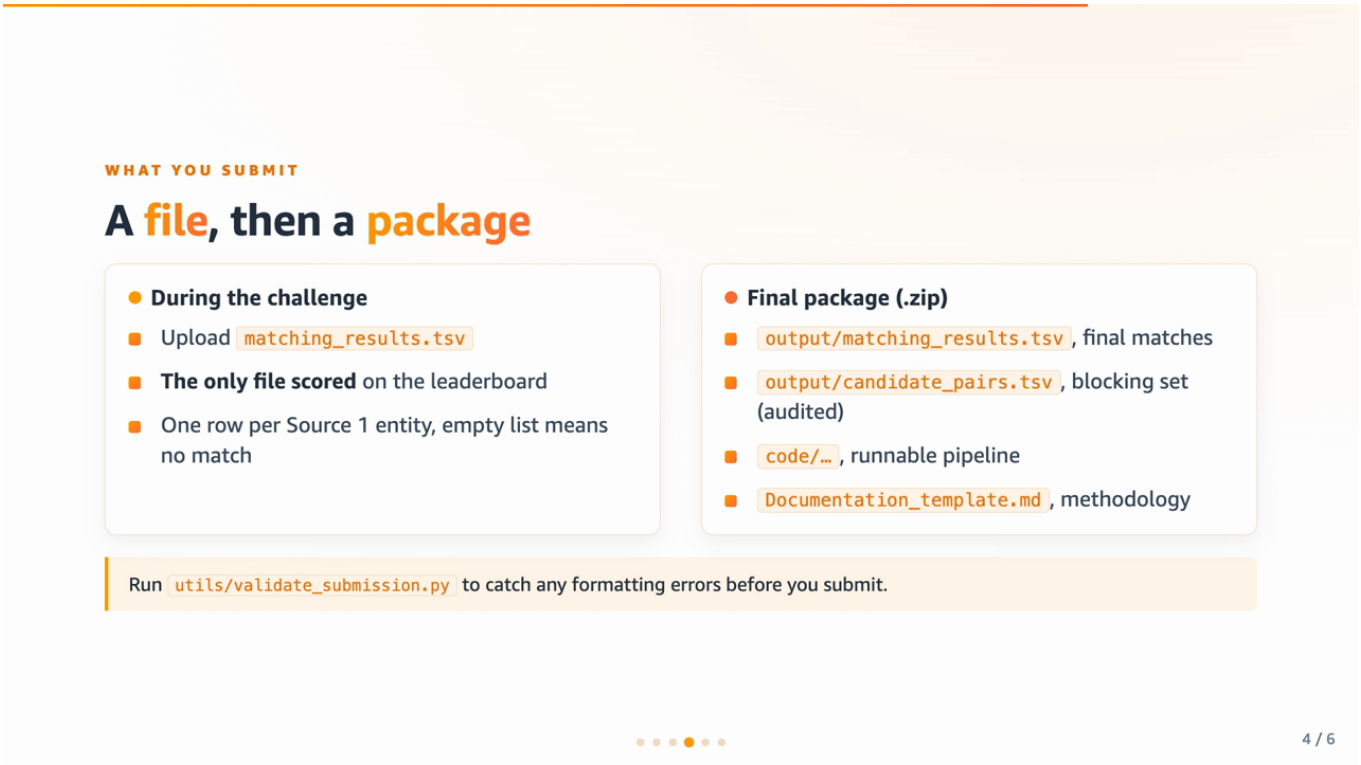
Documentation_template.md methodology

Before submitting, run:

`utils/validate_submission.py`

The purpose of this validation script is to catch formatting errors before submission.

SLIDE 5 OF 6 — JUDGING & TIPS



JUDGING & TIPS

The challenge is scored on macro F_0.5, which is precision-weighted.

The formula shown is:

```
F_0.5 = (1.25 × Precision × Recall) /
```

(0.25 × Precision + Recall)

FALSE MERGES VS MISSES

The slide states:

A false merge costs about twice as much as a miss, so when unsure, do not merge.

This means the scoring setup places greater emphasis on precision than recall.

SINGLETONS

Singletons are Source 1 entities with no match in Source 2 or Source 3.

For a singleton:

• Predict an empty list.

• You earn a full 1.0.

If you predict any match for a singleton:

• You score 0.

BLOCKING AND RECALL

The slide states:

Blocking sets your recall ceiling, so invest there first.

You cannot match a record you never consider.

In other words, if the correct Source 2/Source 3 record is excluded during candidate generation/blocking, the later matching stage cannot recover it.

REGION-SPECIFIC PATTERNS

Account for region-specific patterns in both names and addresses.

NO EXTERNAL DATA LOOKUP

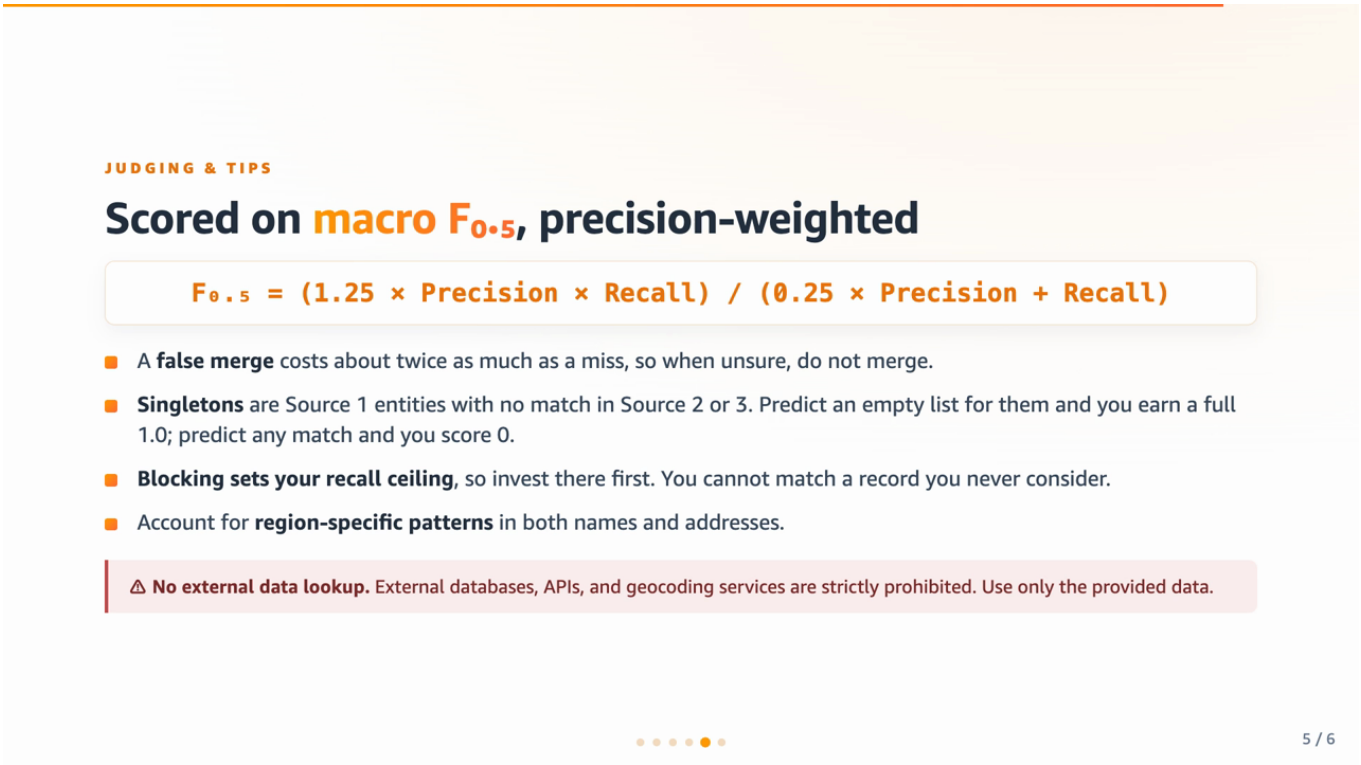
External data lookup is not allowed.

The slide explicitly states:

External databases, APIs, and geocoding services are strictly prohibited.

Use only the provided data.

SLIDE 6 OF 6 — CLOSING



AMAZON ML CHALLENGE 2026

Build something you are proud of, resolve those entities, and enjoy the challenge. We are excited to see what you create.

1. A business signs up. 2. The challenge captures the business name and address. 3. Phone and email exist in the real-world setting, but this challenge uses only name and address. 4. The same real-world business can appear independently in Source 1, Source 2, and Source 3. 5. There is no shared identifier between the sources. 6. Entity resolution must determine which records refer to the same real-world business. 7. Every Source 1 entity can have zero, one, or many matches. 8. Blocking creates candidate groups using a cheap key built from name and address. 9. Blocking is intended to group both true matches and look-alikes. 10. candidate_pairs.tsv contains the resulting candidate/blocking set. 11. The matching model filters those candidates and retains only true matches. 12. matching_results.tsv contains the final matches, with one row per Source 1 entity. 13. The training set contains all three sources plus ground-truth labels in

`train_ground_truth.tsv.`

14. The test set contains the same three sources without labels. 15. A prediction must be produced for every Source 1 entity in the test set. 16. The leaderboard scores matching_results.tsv. 17. All files are TSV files and should be read with sep="\\t". 18. During the challenge, matching_results.tsv is the only file scored on the leaderboard. 19. The final ZIP package additionally contains the audited candidate_pairs.tsv, runnable code, and methodology documentation. 20. utils/validate_submission.py should be run before submission to catch formatting errors. 21. Judging uses macro F_0.5, with the formula shown above. 22. The scoring is precision-weighted. 23. The video states that a false merge costs about twice as much as a miss. 24. Singleton Source 1 entities should receive an empty match list. 25. Blocking determines the recall ceiling because records excluded from the candidate set cannot subsequently be matched. 26. Region-specific name and address patterns should be considered. 27. External databases, APIs, and geocoding services are prohibited. 28. Only the provided data may be used.